

DSP Problem

- 1) 3-tap FIR filter with co-efficients $h_0 = 2, h_1 = 1, h_2 = 3$ & told you the last few i/p samples were $x[n] = 5, x[n-1] = 4, x[n-2] = 1$; could you compute $y[n]$.

Co-efficients : $\underline{h_0}, \underline{h_1}, \underline{h_2}$

$h =$ co-efficients & $x = [] \Rightarrow$ delays.

$$y[n] = \cancel{h_0} x[n] + h_1 x[n-1] + h_2 x[n-2]$$

$$= \underline{2 \cdot 5} \oplus \underline{1 \cdot 4} \oplus \underline{3 \cdot 1}$$

$$= 10 + 4 + 3$$

$$y[n] = 17.$$

- 2) Same filter ($h_0 = 2, h_1 = 1, h_2 = 3$). Now suppose one more sample arrives: $x[n+1] = 6$. What are $x[n+1], x[n], x[n-1]$, relabelled for computing $y[n+1]$ & what's $y[n+1]$.

$$y[n+1] = \cancel{h_0} x[n+1]$$

$$= \cancel{h_0 x[n]} + \cancel{h_1 x[n-1]}$$

$$y[n+1] = h_0 x[n+1] + h_1 x[n] + h_2 x[n-1]$$

$$= 2 x[n+1] + 1 x[n] + 3 x[n-1]$$

$$\cancel{x[n+1]} = -2 \cancel{x[n]} \quad \cancel{x[n]} = -1 \quad \cancel{x[n-1]} = -3$$

$$= 2 \cdot 6 + 1 \cdot 5 + 3 \cdot 4$$

$$= 26$$

$$= 12 + 6 + 12$$

$$= 30.$$

$$2 \times 6 = 12$$

$$1 \times 5 = 5$$

$$3 \times 4 = 12$$

$$\underline{29}$$

Everytime a new sample arrives, the whole window of "current + past samples", shift down by one slot, each value gets multiplied by it's matching co-efficient (always in the same

$$4. \quad y[n] = h_0 x[n] + h_1 x[n-1] + h_2 x[n-2] + h_3 x[n-3]$$

$$= 1.4 + 2.1 + 0.3 + 1.2$$

$$= 4 + 2 + 0 + 2$$

$$= 8$$

$$= x[n+1] + h_0 x[n] + h_1 x[n-1] + h_2 x[n-2] + h_3 x[n-3]$$

$$(n+1) - x = n - 1$$

$$x = n+1 \quad (n+1) - x = n-1$$

$$-x = n-1 - (n+1)$$

$$-x = n-1 - n - 1$$

$$-x = (n-1)$$

$$x = (n-1)$$

$$-x = -2$$

$$x = 2.$$

$x[n+1] \Rightarrow$ is absolute time-index $n+1$

$x[n] \Rightarrow$ is absolute time-index n .

$x[n-1] \Rightarrow$ time-index $n-1$

$x[n-2] \Rightarrow$ time-index $n-2$

Generic FIR formula.

$$y[m] = h_0 x[m] + h_1 x[m-1] + h_2 x[m-2] + h_3 x[m-3]$$

$y[n]$ sub $m = n$.

$$y[n] = h_0 x[n] + h_1 x[n-1] + h_2 x[n-2] + h_3 x[n-3]$$

for $y[n+1]$

$$y[n+1] = h_0 x[n+1] + h_1 x[(n+1)-1] + h_2 x[(n+1)-2] + h_3 x[(n+1)-3]$$

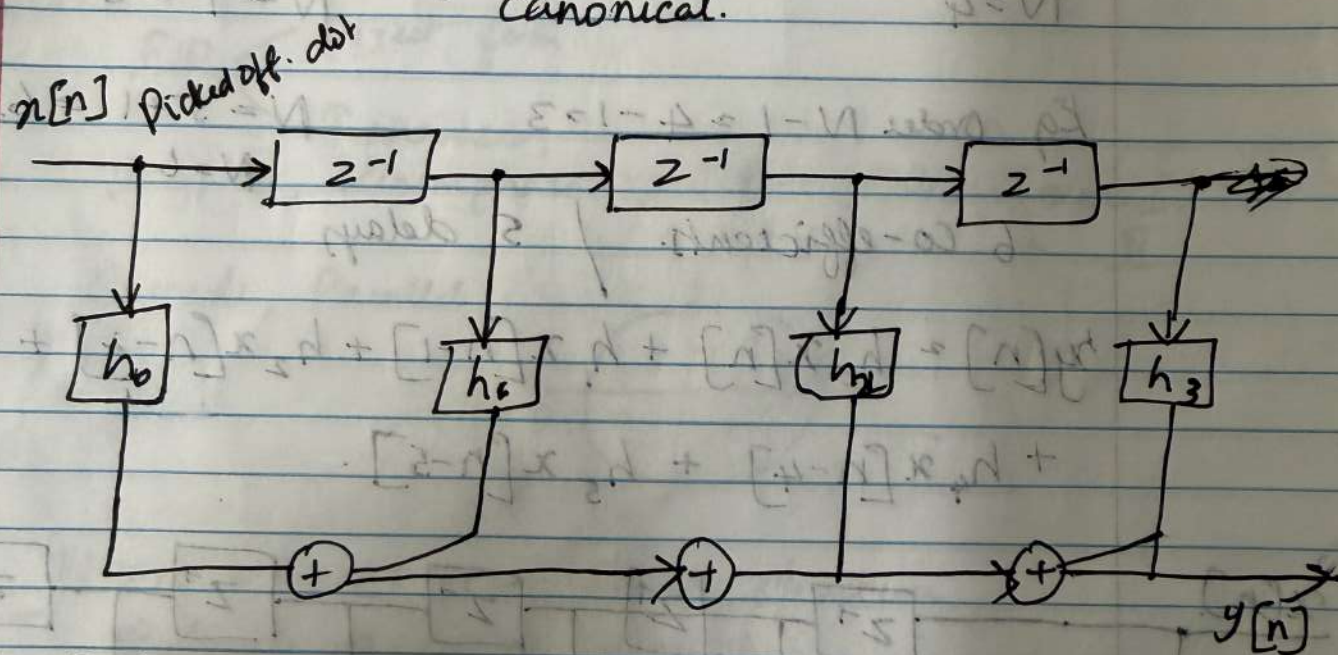
$$= h_0 x[n+1] + h_1 x[n] + h_2 x[n-1] + h_3 x[n-2]$$

$$y[n+2] = h_0 x[n+2] + h_1 x[(n+2)-1] + h_2 x[(n+2)-2] + h_3 x[(n+2)-3]$$

$$y[n] = h_0 x[n] + h_1 x[n-1] + h_2 x[n-2] + h_3 x[n-3]$$

Sample, delay, multiplier, adder, & the sliding window.

Same no. of delays used $\Rightarrow$ Same no. of orders $\Rightarrow$ Canonical.



$h_0 x[n] = h_0 x[n] \Rightarrow$ no delay in above.

$h_1 x[n-1] = h_1 x[n-1] \Rightarrow$ delay before H_1 .

$h_2 x[n-2] = h_2 x[n-2] \Rightarrow$ delay before H_2

$h_3 x[n-3] = h_3 x[n-3] \Rightarrow$ delay b4 H_3 .

Order of an filter $\Rightarrow (N-1)$ Where N is no. of co-efficients.

N - How many co-efficients / Why -1 $\Rightarrow$ not only the co-efficients but its the highest delay.

4-coefficients $\Rightarrow$ "order"

$$N-1 = 4-1 = 3.$$

Step 2: 3 boxes labeled z^{-1} delays.

delays in dia $\Rightarrow$ 3. Order = 3.

4-tap order.

5 order

$$N=4$$

$$N=1=5$$

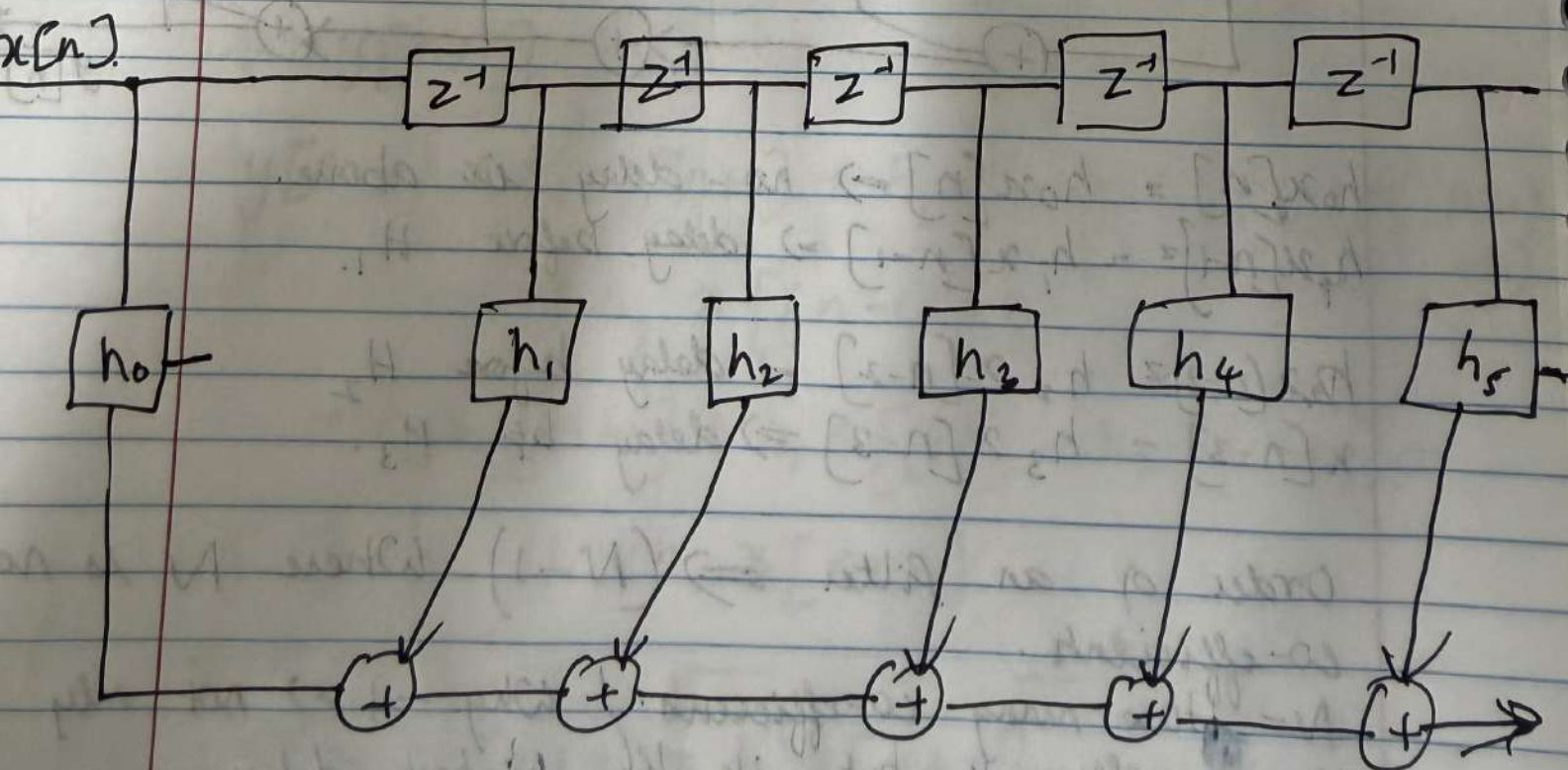
Eg Order $N-1 = 4-1 = 3$

$$N=5+1=6.$$

$$N=6.$$

6 co-efficients. / 5 delays

$$y[n] = h_0 x[n] + h_1 x[n-1] + h_2 x[n-2] + h_3 x[n-3] + h_4 x[n-4] + h_5 x[n-5].$$



$$H(z) = \frac{Y(z)}{X(z)}, \quad Y(z) = H(z) \cdot X(z)$$

FIR

$$H(z) = b_0 + b_1 z^{-1} + b_2 z^{-2} + \dots + b_N z^{-N}$$

IIR

$$H(z) = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2} + \dots + b_N z^{-N}}{1 + a_1 z^{-1} + \dots + a_M z^{-M}}$$

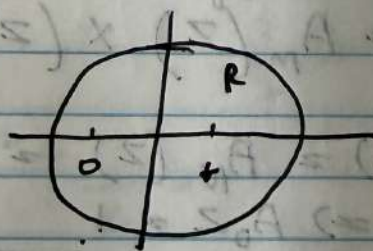
$$1 + a_1(z^{-1}) + \dots + a_M z^{-M}$$

LTI, Causal $\rightarrow$ all poles are inside unit circle.

FIR / Direct form

Transpose, Polyphase, linear phase, IIR - direct form. I

Cascade, Parallel



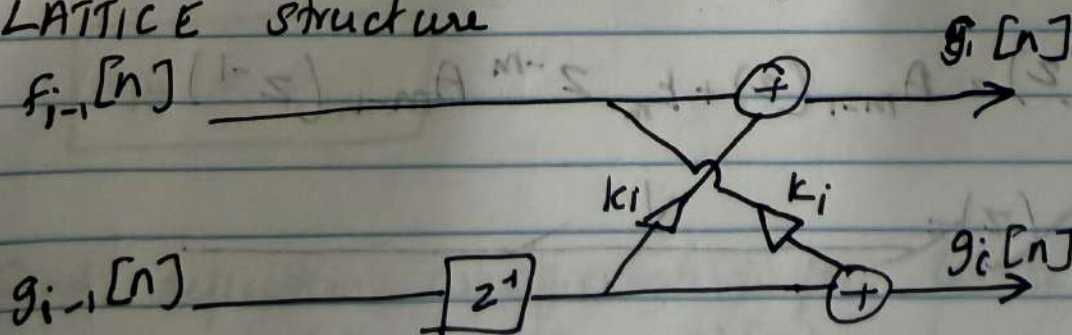
$N > 10 \Rightarrow$ LATTICE format structure

$\Rightarrow$ Simple to ensure stability.

$\Rightarrow$ Modularity.

$\Rightarrow$ low sensitive to co-efficient quantization.

LATTICE structure



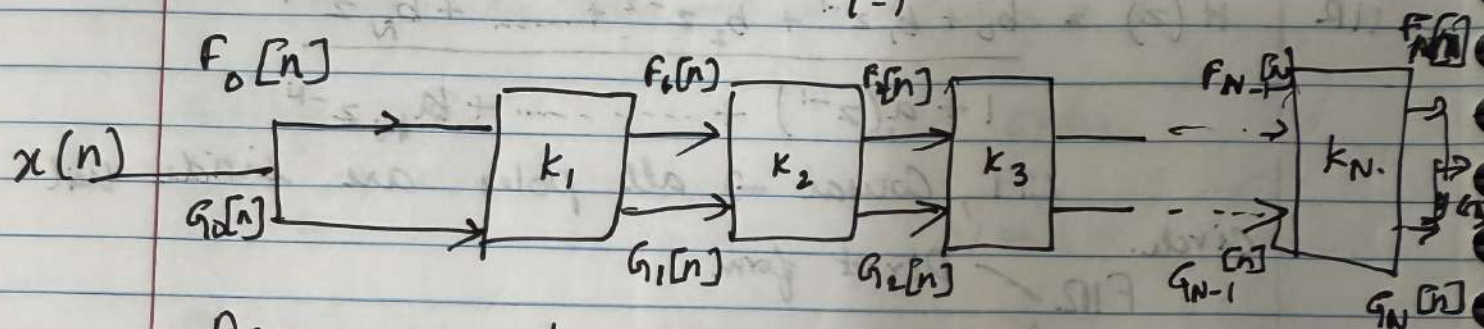
$i/p: f_{i-1}[n], g_{i-1}[n] \quad / \quad \text{Outputs } g_i[n], f_i[n]$

Difference eq: $f_i[n] = f_{i-1}[n] + k_i g_{i-1}[n-1]$

$$g_{i-1}[n] = k_i f_{i-1}[n] + g_{i-1}[n-1]$$

$$F_i[z] = F_{i-1}[z] + k_i z^{-1} G_{i-1}(z)$$

$$G_i[z] = k_i F_{i-1}[z] + z^{-1} G_{i-1}(z)$$

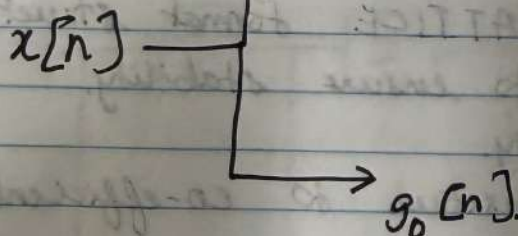


Assume we have $0 \leq m \leq N$.
Transfer function. $x[n] \rightarrow F_N[n]$
If $M = N$

$$F_m(z) = A_m(z) \times (z)$$

$$A_m(z) = A_N(z) = H(z)$$

$$\text{If } m=0 \Rightarrow A_0 z = 1 \rightarrow F_0[n] = y[n]$$



step up.

$$A_m(z) = A_{m-1}(z) + k_m z^{-m} A_{m-1}(z^{-1})$$

$$A_{m-1}(z) = \frac{A_m(z) - k_m z^{-m} A_m(z^{-1})}{1 - k_m^2}$$

step down

$$A_{m-1}(z) = \frac{1}{1 - k_m^2} [A_m(z) - k_m z^{-m} A_m(z^{-1})]$$

1) Given 2nd order FIR LATTICE filter with reflection co-efficients

$$k_1 = 1/2 \text{ and } k_2 = 1/4, \text{ Find } H(z)$$

Assume we are here $0 \leq m \leq N$, k_i 's are called reflectant reflection co-efficients.

$$A_0(z) = 1, A_1(z) = A_0(z) + k_1 z^{-1} A_0(z^{-1})$$

$$A_1(z) = 1 + \frac{1}{2} z^{-1} \cdot 1$$

$$A_2(z) = A_1(z) + k_2 z^{-2} A_1(z)$$

$$= 1 + \frac{1}{2} z^{-2} + \frac{1}{4} z^{-2} \left(1 + \frac{1}{2} z \right)$$

$$A_2[z] = 1 + \frac{1}{2} z^{-2} + \frac{1}{4} z^{-2} \left(1 + \frac{1}{2} z \right)$$

$$A_2[z] = 1 + \frac{z^{-2}}{2} + \frac{z^{-2}}{4} + \frac{z^{-1}}{4} = H(z)$$

We know $H(z) = A_N(z)$. Let $A_N(z) = 1 + a_1 z^{-1} + a_2 z^{-2} + \dots + a_N z^{-N}$

$$A_m(z) = 1 + a_1 z^{-1} + a_2 z^{-2} + \dots + a_m z^{-m}$$

$$\boxed{k_m = a_m}$$

2) Find reflection co-efficients for lattice realization an FIR filter with $H(z) = 1 - \frac{1}{2} z^{-2}$

$$A_2(z) = H(z) = 1 - \frac{1}{2} z^{-2} \quad \left| \quad a_2 = k_2 = -\frac{1}{2} \right.$$

$$A_1(z) = \frac{1}{1 \mp 1/4} [A_2[z] - k_2 z^{-2} A_1[z]]$$

$$= \frac{1}{1 \mp 1/4} \left[A_2[z] + \frac{1}{2} z^{-2} A_1[z] \right]$$

$$= \frac{1}{3/4} \left[\cancel{A_2[z]} - \frac{1}{2} \right]$$

$$= \frac{1}{3/4} \left[\left(1 - \frac{1}{2} z^{-2} \right) + \frac{1}{2} z^{-2} \left(1 - \frac{1}{2} z^{-2} \right) \right]$$

$$A_1(z) = \frac{1}{3/4} \left[\cancel{1 - \frac{1}{2} z^{-2}} + \frac{1}{2} + \frac{1}{2} z^{-2} - \frac{1}{4} z^{-4} \right]$$

$$\frac{1}{3/4} \left(\frac{3}{4} \right) = 1.$$

$$= 1 + 0z^{-1}$$

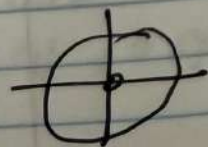
$$\swarrow a_1 = k_1 = 0.$$

FIR, DIR & lattice. $\Rightarrow$ With FPGA

All pole IIR.

$$H(z) = \frac{1}{1 + a_1 z^{-1} + \dots + a_N z^{-N}}$$

all zeroes located at $z=0$.

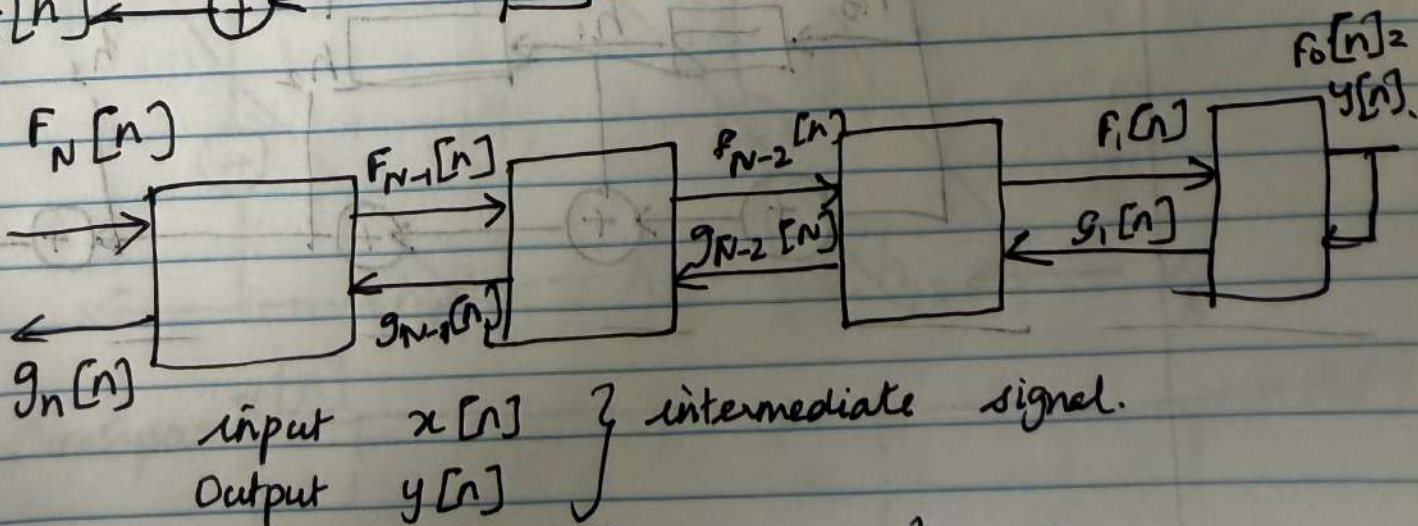
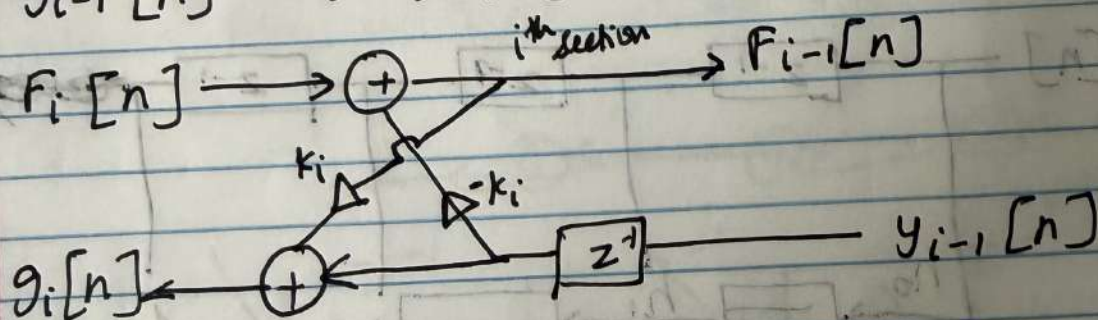


$$H(z) = \frac{1}{A_N(z)} \Rightarrow A_N(z) = 1 + a_1 z^{-1} + \dots + a_N z^{-N}$$

$$f_{i-1}[n] = f_i[n] - k_i g_i[n]$$

$$f_{i-1}[n] = f_i[n] - k_i g_i[n]$$

$$g_{i-1}[n] = k_i f_{i-1}[n] + g_{i-1}[n-1]$$



$$\begin{aligned} f_0[n] &= x[0] \\ g_0[n] &= x[n] \\ f_N[n] &= y[n] \end{aligned}$$

$$\begin{aligned} f_0[n] &= y[n] \\ g_0[n] &= y[n] \\ f_N[n] &= x[n] \end{aligned} \left\{ \begin{aligned} & \frac{x(z)}{y(z)} = A_N(z) \\ & y(z) = x(z) \end{aligned} \right.$$

$$\frac{x(z)}{y(z)} = A_N z$$

$$F_N(z) = x(z) = A_N(z) \cdot y(z)$$

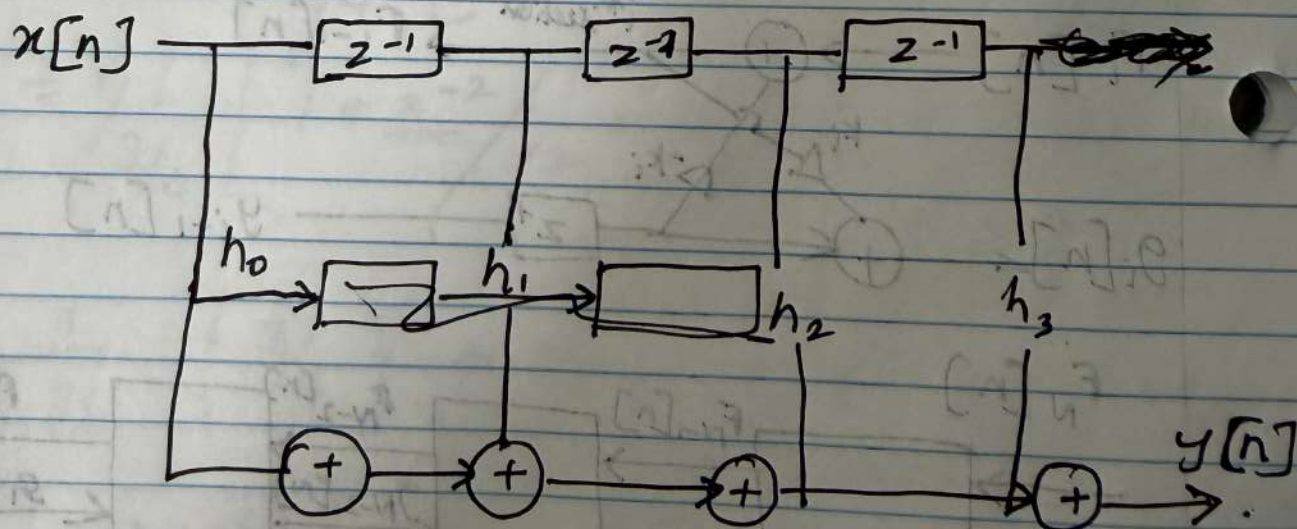
$$\text{if } m=N, \frac{x(z)}{y(z)} = A_N(z) \Rightarrow H(z) = \frac{1}{A_N(z)}$$

if $m=0$

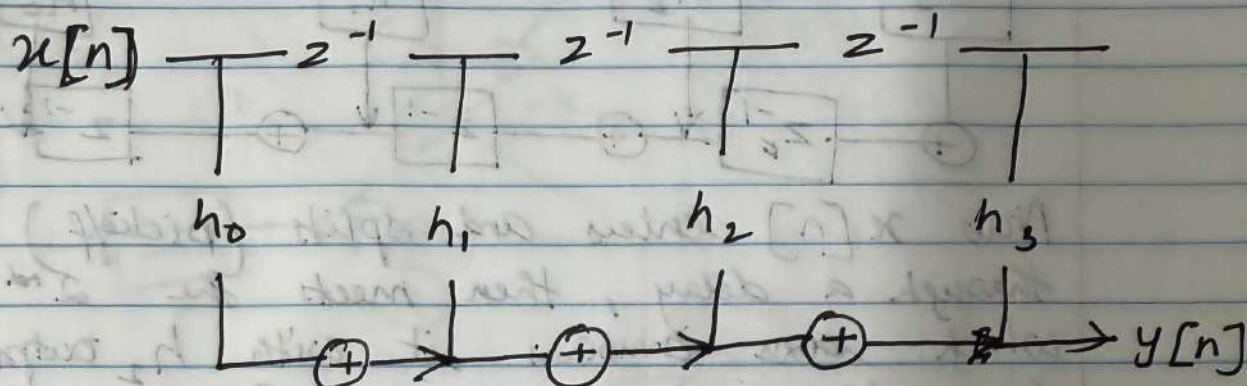
$$F_0[n] = y[n]$$

$$g[n]$$

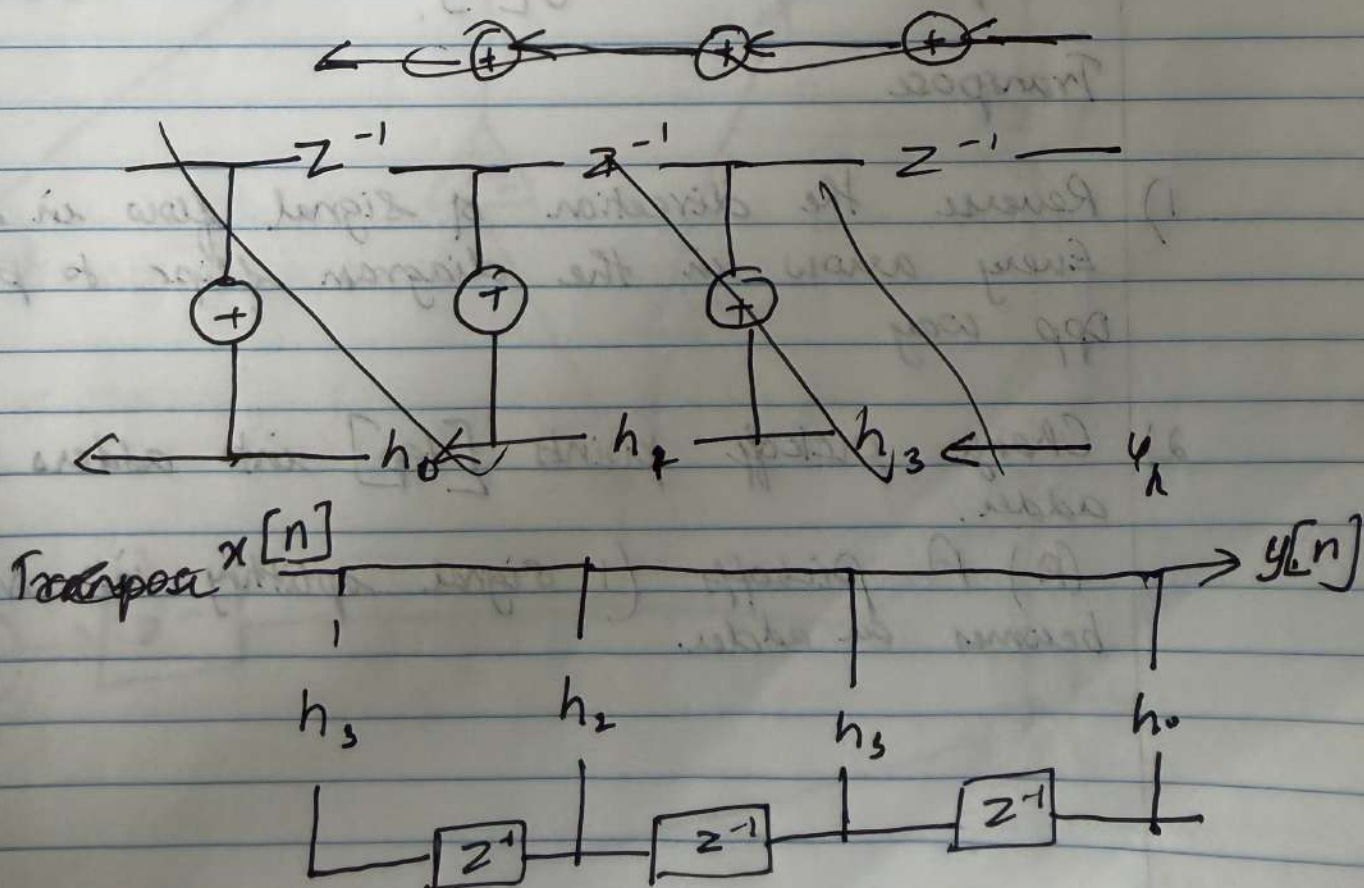
$$A_0(z) = 1$$

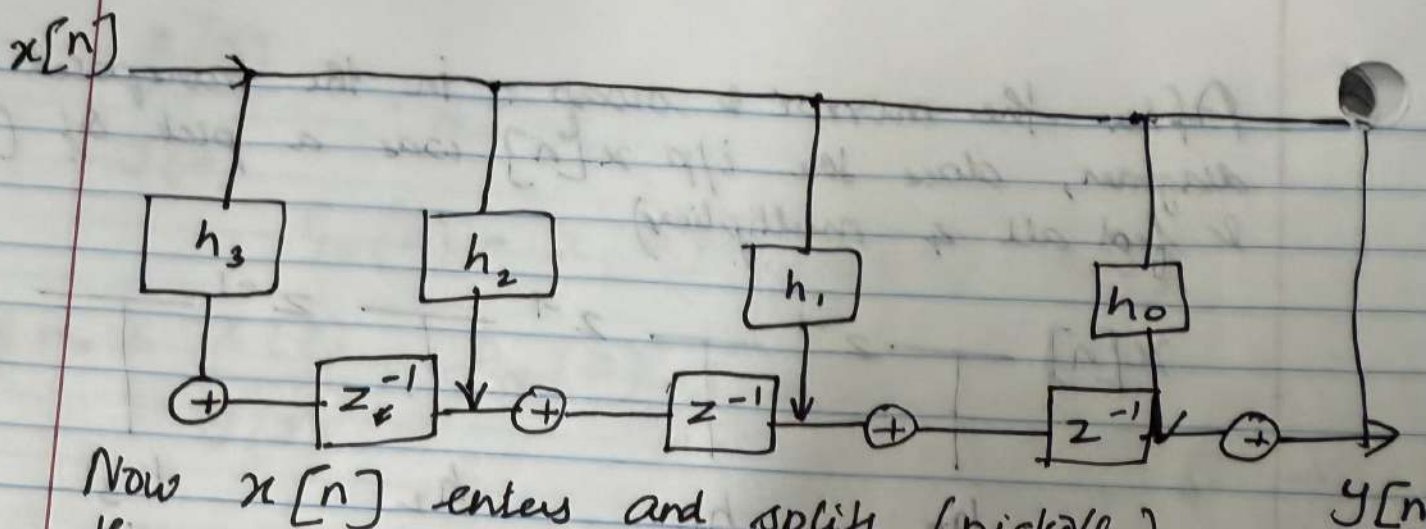


After the mirror & swap : in the transposed diagram, does the i/p $x[n]$ was a pick off (it split & feed all 4 multipliers)



h_0, h_1, h_2 & $h_3 \Rightarrow$ Multipliers.



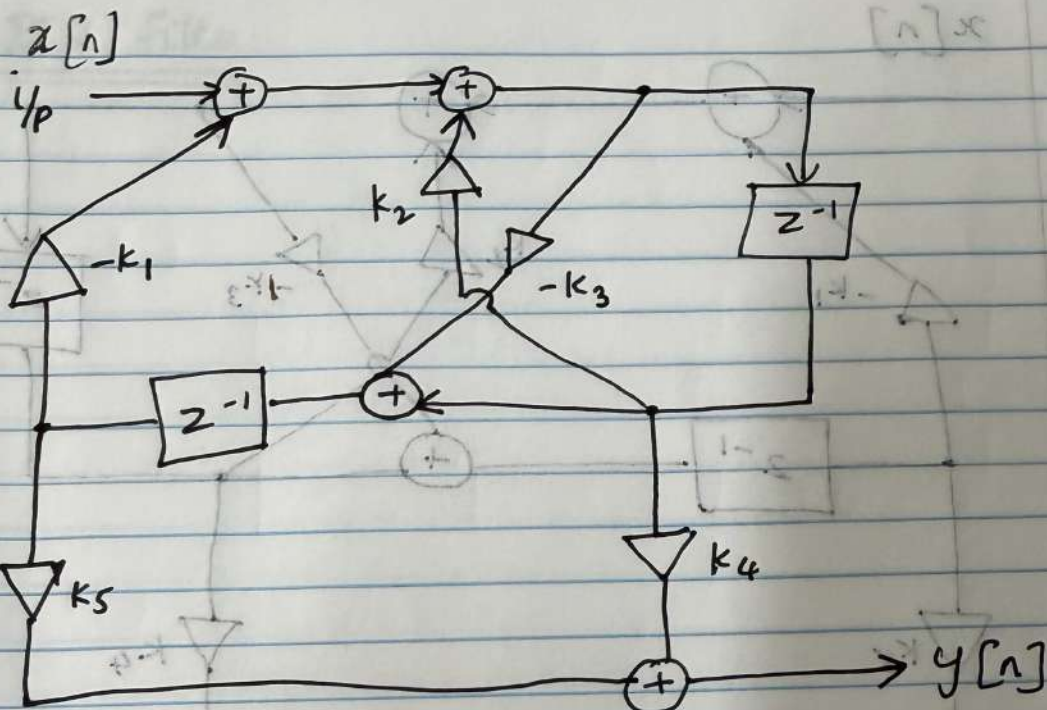


Now $x[n]$ enters and splits (pickoff) through a delay, then meets the 2nd adder, which ~~sum~~ combines it with h_2 output (from the next pickoff of $x[n]$ along the top). That sum passes through another delay, hits the 3rd adder combines with h_1 's o/p, delay again hits the 4th adder. combines with h_0 o/p & the final sum $\rightarrow y[n]$.

Transpose

- 1) Reverse the direction of signal flow in all paths. Every arrow in the diagram flips to point the opp way.
- 2) Change pickoff points [tap] into adders & into (A) A pickoffs (1 signal splitting into many) $\rightarrow$ becomes an adder.

Homework 1



Transpose

